HDOC 47

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BATCH 15

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1. Algorithm

1. Start.
2. Read the size `n` of the sorted array.
3. Read the array elements in ascending order.
4. Read the target value.
5. Pass the array to the `ceil_index()` function using a pointer.
6. Find the first element greater than or equal to the target.
7. Return its index. If no such element exists, return `-1`.
8. Display the ceiling element and its index, or `-1`.
9. Stop.

2. C Program

```c
#include <stdio.h>

int ceil_index(int *arr, int n, int target)
{
    int i;

    for (i = 0; i < n; i++)
    {
        if (*(arr + i) >= target)
        {
            return i;
        }
    }

    return -1;
}

int main()
{
    int n, target, i, index;

    printf("Enter array size: ");
    scanf("%d", &n);

    if (n <= 0)
    {
        printf("Invalid array size\n");
        return 0;
    }

    int arr[n];

    printf("Enter sorted array elements: ");
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter target: ");
    scanf("%d", &target);

    index = ceil_index(arr, n, target);

    if (index == -1)
    {
```

```
        printf("-1\n");
    }
    else
    {
        printf("Ceil of %d: %d at index %d\n",
               target, arr[index], index);
    }

    return 0;
}
```

3. Sample Output

Example 1

```
Enter array size: 6
Enter sorted array elements: 5 7 7 8 8 10
Enter target: 6
Ceil of 6: 7 at index 1
```

Example 2

```
Enter array size: 6
Enter sorted array elements: 5 7 7 8 9 10
Enter target: 8
Ceil of 8: 8 at index 3
```

Example 3

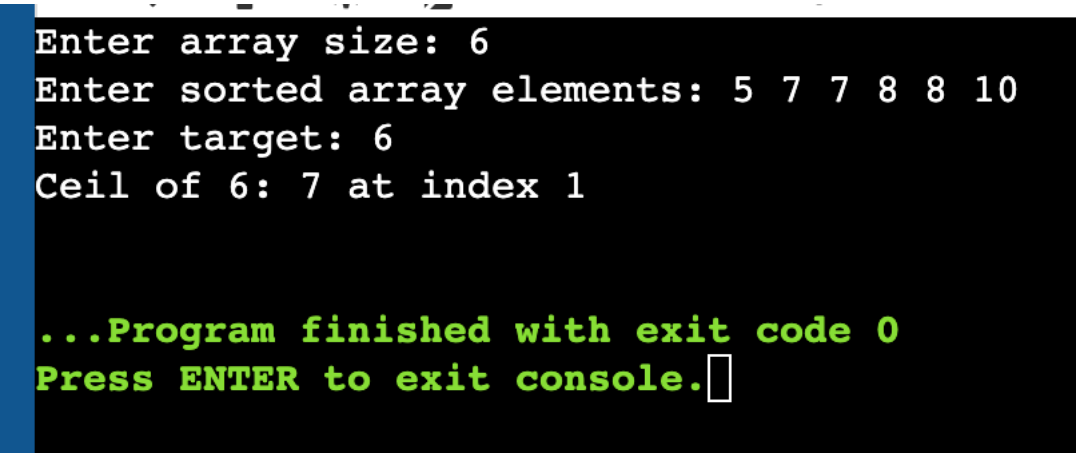
```
Enter array size: 6
Enter sorted array elements: 5 7 7 8 8 10
Enter target: 11
-1
```

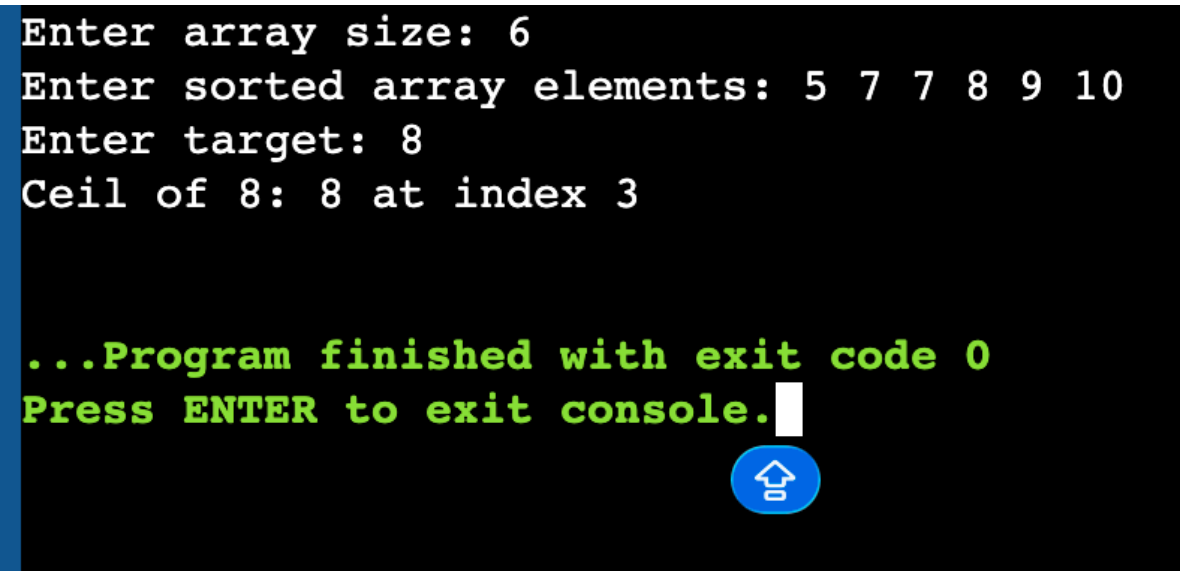
4. Test Cases

Test	Target	Expected output
1	6	Ceil of 6: 7 at index 1
2	8	Ceil of 8: 8 at index 3
3	11	-1
4	5	Ceil of 5: 5 at index 0
5	9	Ceil of 9: 10 at index 5
6	7	Ceil of 7: 7 at index 1

5. Result

The C program successfully finds the ceiling element of a given target in a sorted integer array using pointers and returns the index of its first occurrence. If no element is greater than or equal to the target, it displays `-1`.





```
Enter array size: 6
Enter sorted array elements: 5 7 7 8 8 10
Enter target: 11
-1


...Program finished with exit code 0
Press ENTER to exit console.
```

